

Problem 1 - Parts Method:

$$1 + 4 = 5 \quad (\text{Total parts})$$

$$100 \div 5 = 20 \quad (\text{Each part})$$

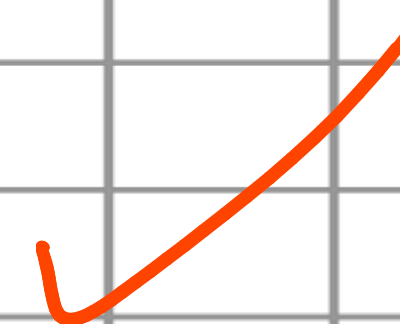
$$a = 1 \times 20$$

$$= 20$$

$$b = 4 \times 20$$

$$= 80$$

$$a = 20 ; \quad b = 80$$



Problem 2-Parts Method:

$$5 + 3 = 8 \quad (\text{Total parts})$$

$$7 \div 8 = 9 \quad (\text{Each part})$$

$$a = 9 \times 5$$

$$= 45$$

$$b = 9 \times 3$$

$$= 27$$

$$a = 45; \quad b = 27$$



Problem 3-Parts Method:

$$2 + 3 = 5 \quad (\uparrow \text{Total parts})$$

$$150 \div 5 = 30 \quad (\text{Each part})$$

$$a = 30 \times 2$$

$$= 60$$

$$b = 30 \times 3$$

$$= 90$$

$$a = 60; \quad b = 90$$



Problem 4-Parts Method:

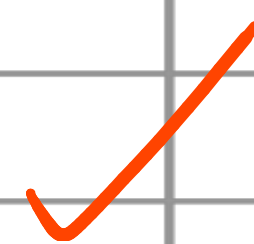
$$a + b = 1000; \quad a : b = 7 : 3$$

$$\begin{aligned} \text{Total parts} &= 7 + 3 \\ &= 10 \end{aligned}$$

$$\begin{aligned} \text{Value per part} &= \frac{1000}{10} \\ &= 100 \end{aligned}$$

$$\begin{aligned} a &= 100 \times 7 \\ &= 700 \end{aligned}$$

$$\begin{aligned} b &= 100 \times 3 \\ &= 300 \end{aligned}$$



Problem 5 - Parts Method:

$$a + b = 45; \quad a : b = 4 : 5$$

$$\begin{aligned} \text{Total parts} &= 4 + 5 \\ &= 9 \end{aligned}$$

$$\begin{aligned} \text{value per part} &= 45 \div 9 \\ &= 5 \end{aligned}$$

$$\begin{aligned} a &= 5 \times 4 \\ &= 20 \end{aligned}$$

$$\begin{aligned} b &= 5 \times 5 \\ &= 25 \end{aligned}$$



Problem 6-Substitution:

$$S \text{ ug a } \overset{(s)}{\leftarrow} r : f l o \overset{(F)}{\leftarrow} v = 2 : 7 ;$$

$$S + F = 450g$$

$$S = \frac{2}{7} \times F$$

$$S + F = 450$$

$$\frac{2}{7} \times F + F = 450$$

$$\frac{2}{7} \times F + \frac{7}{7} \times F = 450$$

$$\frac{9}{7} \times F = 450$$

$$\frac{9}{7} \times \frac{7}{9} \times F = 450 \times \frac{7}{9}$$

$$F = \underline{350}$$

$$F = 350g$$

$$S = 450g - 350g$$

$$= 100g$$

$$\begin{array}{r} 350 \\ \times 7 \\ \hline 2450 \end{array}$$

$$\begin{array}{r} 350 \\ 9 \overline{) 3150} \\ \underline{27} \\ 045 \\ \underline{45} \\ 000 \\ \underline{00} \\ 0 \end{array}$$

$$\text{Total parts} = 2 + 7 = 9$$

$$\text{Each part} = \frac{450}{9} = 50g$$

$$S = 2 \times 50 = 100g$$

$$F = 7 \times 50 = 350g$$

Problem 7:

Board length = 120 cm

Ratio: $3:5$

$$\Rightarrow a + b = 120;$$

(a = smaller piece)
(b = larger piece)

$$\left| \begin{array}{l} \frac{a}{b} = \frac{3}{5} \\ a = \frac{3}{5} \times b \end{array} \right.$$

$$\frac{3}{5} \times b + b = 120$$

$$\Rightarrow \frac{3}{5} \times b + \frac{5}{5} \times b = 120$$

$$\Rightarrow \frac{8}{5} \times b = 120$$

$$b = 120 \times \frac{5}{8}$$

$$= \frac{600}{8}$$

$$= 75 \text{ cm}$$

$$\Rightarrow a = 120 - 75$$

$$= 45 \text{ cm}$$



$$8 \overline{) 600} \text{ C75}$$

$$\begin{array}{r} 600 \\ \underline{000} \\ 000 \end{array}$$

$$\begin{array}{r} -1 \quad -1 \\ 1 \quad 2 \quad 0 \\ \underline{ 7 \quad 5} \\ 0 \quad 4 \quad 5 \end{array}$$

$$45 \overline{) 75} \text{ C1}$$

$$\begin{array}{r} 4 \quad 5 \\ \underline{3 \quad 0} \end{array} \overline{) 45} \text{ C1}$$

$$\begin{array}{r} 7 \quad 5 \\ 8 \overline{) 600} \\ \underline{56} \\ 040 \\ \underline{40} \\ 00 \end{array}$$

$$\begin{array}{r} 3 \quad 0 \\ 15 \overline{) 30} \text{ C2} \\ \underline{30} \\ 00 \end{array}$$

Problem 8: Substitution:

$$c h i c k e n s = a$$

$$c o w s = b$$

$$a + b = 121; a : b = 10 : 1$$

$$a = 10 \times b$$

$$a + b = 121$$

$$\Rightarrow 10 \times b + b = 121$$

$$\Rightarrow 11 \times b = 121$$

$$\Rightarrow \frac{11 \times b}{11} = \frac{121}{11}$$

$$\Rightarrow b = 11$$

$$\Rightarrow a = 10 \times b$$

$$= 10 \times 11$$

$$= 110$$



Problem 9:

$$a + b = 210,$$

$$a \text{ and } a = 25\% \times b$$

$$\Rightarrow a = \frac{25}{100} \times b$$

$$\Rightarrow b = \frac{840}{5}$$

$$\Rightarrow b = 168$$

$$\Rightarrow a = \frac{1}{4} \times b$$

$$\Rightarrow a = 210 - 168$$

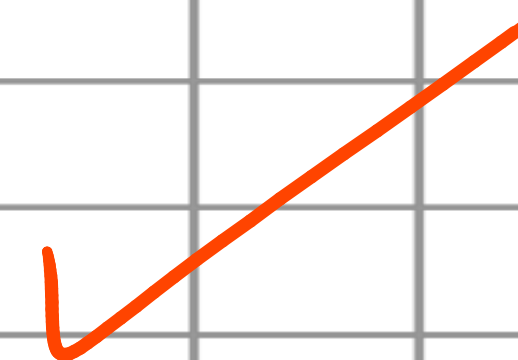
$$\Rightarrow \frac{1}{4} \times b + b = 210$$

$$= 42$$

$$\Rightarrow \frac{1}{4} \times b + \frac{4}{4} \times b = 210$$

$$\Rightarrow \frac{5}{4} \times b = 210$$

$$\Rightarrow \frac{5}{4} \times \frac{4}{5} \times b = 210 \times 4 \div 5$$



$$25 \overline{)100} (4$$

$$\begin{array}{r} 100 \\ \underline{000} \end{array}$$

$$\begin{array}{r} 210 \\ - 168 \\ \hline 042 \end{array}$$

$$\begin{array}{r} 210 \\ \times 4 \\ \hline 840 \end{array}$$

Problem 10:

$$a + b + c = 180$$

$$a : b : c = 2 : 3 : 4$$

Using parts method

Total parts =

$$\begin{array}{l} \text{Each part} \rightarrow 2 + 3 + 4 = 9 \\ \text{Total} \rightarrow 180 \div 9 = 20 \end{array}$$

$$a = 20 \times 2$$

$$= 40$$

$$b = 20 \times 3$$

$$= 60$$

$$c = 20 \times 4$$

$$= 80$$



$$\begin{array}{r} 80 \\ + 60 \\ + 40 \\ \hline 180 \end{array}$$